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FUNCTIONAL ROLE OF MITOCHONDRIAL REACTIVE OXYGEN SPECIES IN PHYSIOLOGY

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ABSTRACT

The major energy generator in the cell – mitochondria produce reactive oxygen species as a by-product of a number of enzymatic reactions and the production of ATP. Emerging evidence suggests that mitochondrial ROS regulate diverse physiological parameters and that dysregulated ROS signalling may contribute to a development of processes which lead to human diseases. ROS produced in mitochondrial enzymes are triggers of monoamine-induced calcium signal in astrocytes, playing important role in physiological and pathophysiological response to dopamine. Generation of ROS in mitochondria leads to peroxidation of lipids, which is considered to be one of the most important mechanisms of cell injury under condition of oxidative stress. However, it also can induce activation of mitochondrial and cellular phospholipases that can trigger a variety of the signals – from activation of ion channels to stimulation of calcium signal. Mitochondria are shown to be the oxygen sensor in astrocytes, therefore inhibition of respiration by hypoxia induces ROS production which leads to lipid peroxidation, activation of phospholipase C and induction of IP_3-mediated calcium signal. Propagation of astrocytic calcium signal stimulates breathing activity in response to hypoxia. Thus, ROS produced by mitochondrial enzymes or electron transport chain can be used as a trigger for signalling cascades in central nervous system and deregulation of this process leads to pathology.

Introduction

Although mitochondria have multiple important functions, these organelles are mainly high efficient engine for conversion of the products of glucose or fatty acid metabolism into

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universal form of energy which is accepted by all types of cells – ATP. As in every highly efficient engine, the process of ATP production in the oxidative phosphorylation coupled with the mitochondrial respiration in the electron transport chain is having a leakage of energy in the [1]form of electrons. These outgoing electrons can hit the nearby situated molecules and ultimately produce free radicals. Mitochondria are using most of the oxygen taken up by the cell and considering this oxygen-rich environment, the production of Reactive Oxygen Species (ROS) is inevitable. Electrons escaping the electron transport chain in the first instance generate ROS mainly in the form of super oxide radical $O_2^{\cdot-}$. This superoxide is converted to hydrogen peroxide [2, 3]; [4].

Compared to the activated enzymes such as NADPH oxidases or xanthine oxidase mitochondria produce less ROS, but in contrast to any other enzymatic and non-enzymatic ROS producer, this organelles are able to generate free radicals continually [5]. Considering the high activity and toxicity of some forms of ROS, our cells (and particularly the mitochondria) in the time of evolution have not only developed effective defence systems against oxidative damage but also adopted ROS to play signalling and regulatory role in physiological processes.

The rate of production and the level of ROS produced by mitochondria can be modulated by a number of factors that render this system to be an ideal regulatory instrument in the cellular homeostasis and to be a messenger in the signal transduction cascades. Additionally, some misfolded proteins that play role in neurodegeneration (including alpha-synuclein, beta-amyloid and tau) could have an effect on mitochondrial function [6, 7] and ROS production; moreover, oligomeric form of alpha-synuclein in combination with metal ions can induce superoxide and hydrogen peroxide production [8, 9].

In the following review we discuss the physiological role of ROS produced in the mitochondria and how deregulation of this processes can lead to a development of pathology and become trigger of cell death (which is as well one of the multiple mitochondrial functions).

Mitochondrial ROS producers

In the majority of publications under the term “mitochondrial ROS” the authors mean the ROS produced in the electron transport chain (ETC) of the mitochondria. However, ROS in mitochondria can be formed by enzymatic action of numerous enzymes including monoamine oxidase (MAO) and cytochrome b5 reductase (Cb5R) located on the outer mitochondrial membrane, as well as glycerol-3-phosphate dehydrogenase and in some cell types or various cytochrome P450 enzymes located in the inner mitochondrial membrane. There are also several matrix enzymes and complexes including aconitase, pyruvate dehydrogenase (PDH), and α-ketoglutarate dehydrogenase (αKGDH) that can generate superoxide [10]. Superoxide can rapidly dismutate spontaneously or enzymatically with the help of manganese superoxide dismutase in mitochondrial matrix or by Cu,Zn-superoxide dismutase in the intermembrane space. The formed freely permeable H_2O_2 can be further degraded by enzymes such as catalase (CAT), glutathione peroxidase (GPx), and peroxiredoxin 3 [11].

Production of ROS in the ETC

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The main source of reactive oxygen species in the mitochondria is the electron leak from the electron transport chain (ETC). Depending on the direction of the electron "escape" from ETC, ROS (mainly superoxide anion; $O_2^{\cdot-}$) is produced in the matrix or between inner and outer membrane of mitochondria (Figure 1). Electron transport between the complexes of the electron transport chain is limited by the availability of substrates for this complexes (NADH and succinate), but also on the rate of production of substrates in the TCA cycle, the rate of consumption of ATP and/or on availability of the final electron acceptor (oxygen) of the electron transport. However, ROS production that is linked to the topology of the electron carriers and their functional integration is determined by the rate constants of the enzymatic reactions of the complexes of the electron transport chain. Moreover, ROS production is dependent on the values of the mitochondrial membrane potential [12, 13]. The leak of electrons can be induced by inhibition of complex I and III with specific inhibitors (e.g. rotenone and antimycin A), that lead to a higher rate (up to 300 % of the basal rate) of reactive oxygen production in the mitochondria [14, 15]. Inhibition of respiration and decrease of mitochondrial membrane potential by anoxia or hypoxia can stimulate superoxide production in the same way but only for short time [16, 17]. High mitochondrial membrane potential also increases a rate of ROS production in ETC and it can be elevated by application of mitochondrial substrates such as pyruvate or succinate [18, 19]. Mild uncoupling with low doses of FCCP or physiological uncouplers significantly reduces production of superoxide and for long time is used as a basis for cell protective strategy [20, 21]. Although the use of chemical uncouplers reduced mitochondrial ROS production, the decrease of oxidative stress in general may be limited because FCCP on the other hand can maximise superoxide production in activated NADPH oxidase [22]. Thus, the rate of ROS production in ETC can be manipulated by changing the mitochondrial membrane potential, using inhibitors and substrates of the mitochondrial enzymatic complexes.

ROS production in the matrix of mitochondria – TCA

It has been confirmed that ROS production in mitochondria takes place as well outside the electron transport chain: isolated α-ketoglutarate dehydrogenase (KGDHC; the rate-limiting enzyme of the TCA cycle) or pyruvate dehydrogenase complexes (PDHC) are producing superoxide or hydrogen peroxide in the absence of ETC complex inhibitors [23]. Interestingly, despite generating ROS, α-ketoglutarate dehydrogenase complex is on the same time very sensitive to oxidative stress, e.g. peroxynitrite inhibited purified KGDHC activity in a dose-dependent manner and its activity could be restored by GSH [24]. KGDHC can be also inactivated by hydrogen peroxide in bovine cardiac mitochondria, fibroblasts or N2 cells (Figure 1). Moreover, inhibition of KGDHC by hydrogen peroxide limits the amount of NADH available to the complex I from the respiratory chain [25, 26]. Reduced activity of brain α-ketoglutarate dehydrogenase complex occurs in a number of neurodegenerative diseases like Parkinson's disease and Alzheimer's disease, where oxidative stress is thought to be central to the pathophysiology of these diseases [27-29]. Further, in mitochondria the succinate dehydrogenase (SDH) complex as well is capable of superoxide production in the absence of an electron acceptor [30].

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ROS produced at the outer membrane of mitochondria

One of the most effective pharmacological targets in neurology is the mitochondrially located (outer membrane) family of enzymes monoamine oxidase (MAO; Figure 1). MAO catabolises dietary monoamines through oxidative deamination, producing aldehydes and H_2O_2 [31]. ROS production in the form of hydrogen peroxide can be stimulated by catabolism of monoamines from food or generally in the homeostasis of neurotransmitters and neuromodulators – dopamine, serotonin and adrenaline [32]. There are possible intercommunications between ROS from MAO and ETC – for instance, deficiency in MAO-A reveals protection against inhibitors of mitochondrial complexes I, III and IV [33].

Regulatory role of mitochondrial ROS

Mitochondrial ROS as activators of phospholipases

Phospholipases play housekeeping and signalling roles in the cells. It has been demonstrated that ROS production could activate phospholipases [34]. This activating effect has been shown for different forms of ROS including the product of oxidative stress – lipid peroxidation [35-37]. Lipid peroxidation is affecting different isoforms of phospholipase A in mitochondria, cytosol and lysosomes. Stimulating effect of lipid peroxidation has been also shown for phospholipase D [38] and, importantly for phospholipase C [39, 40]. Several mechanisms of phospholipase activation by lipid peroxidation were suggested, including changes of membrane density and structural changes in the phospholipid bilayer. We have recently shown that only lipids that are oxidized can be used as a substrate for phospholipase C [40]. This is a natural process by which damages in the membranes are repaired and further imply a possibly important role of PLC as a part of the cellular antioxidant defence system and a role of ROS and lipid peroxidation in calcium signalling by producing IP_3. In our view, this is a very important observation, because in many studies lipid peroxidation is used as a marker of oxidative stress without considering formations of oxidised lipids as a natural physiological reaction. Also, in confirmation of this, mutation and loss of phospholipase activity lead to increased lipid peroxidation in mitochondria and cytosol and can be a trigger for neurodegeneration [41]. Increased levels of lipid peroxidation is described for various types of pathology [42] playing an important role in cytochrome c release from mitochondria in initiation of cell death [43].

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Role of ROS in signal transduction

It has been observed that application of low doses of hydrogen peroxide to the cells induces calcium signal [40, 44, 45]. Considering this, activation of endogenous ROS (and particularly H_2O_2) formation in mitochondria should play an important role in the initiation of signal transduction. Thus, dopamine-induced calcium signal in astrocytes is due to production of hydrogen peroxide in MAO and it can be blocked by molecular or pharmacological inhibition of this enzyme or by application of antioxidants [46]. Generation of hydrogen peroxide in

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MAO stimulates lipid peroxidation, activation of phospholipase C and generation IP_3-dependent calcium signal [46] confirming the important role of mitochondrial ROS in the signal transduction in CNS. We assume that other monoamines (including adrenaline) can produce similar mechanism of calcium signal in astrocytes or other cell types.

An important role for ROS has been shown for modulation of various ion channels (BK, KCN, ERG, TRP, etc.) and consequently for the regulation of the neuronal and cardiomyocyte excitability [47-51]. Changes in the intracellular redox milieu can affect many physiological processes, e.g. the gating properties of ion channels and the activity of certain transport mechanisms and therefore it could play an essential role in the cellular signal transduction process in a variety of systems. Cysteine or methionine residues on diverse (principal or auxiliary) channel and receptor protein subunits could act as redox–sensitive switches and thus superoxide and hydrogen peroxide could modulate the activity of these proteins [52-54]. For example, ryanodine receptor (RyR) and Ca^{2+} channels activity from the sarcoplasmic reticulum (SR) in the heart can be modulated by both ROS and RNS and therefore have a profound effect on the proper cardiac function and on the development of a pathophysiological condition (cardiac failure) [55].

In fact ROS exert general signalling role in normal physiological processes as in many studies application of antioxidants to a system at rest or non-pathology actually attenuates the normal function of neuronal or cardiac cell excitability for example [47, 56]

Hence, mitochondrial ROS (and RNS), produced in this highly dependent on the redox potential milieu, are very likely to be involved in the regulation of mitochondrial ion channels and transporters [57].

Physiological function of mitochondrial ROS – regulation of breathing

We have recently discovered a physiological role for astrocytes in regulation of breathing, appointed till recently only for the glomus cells from the carotid body [58]. Astrocytes from the brain stem and also from the cortex were able to sense directly even slight decrease in partial pressure of atmospheric oxygen (from 21% to 15% ppO_2). Astrocytes from either primary culture, organotypic culture, acute brain slice or somatosensory cortex *in vivo* in anesthetised animal were able to respond to argon-induced hypoxia in physiological variation range with rise in cytosolic calcium under the form of sustained $[Ca^{2+}]_{i,}$ elevation, calcium oscillations or combination of both. More important, the threshold for astroglial activation of a calcium signal in the cortex lies at about 10 mmHg lower than the value for activation of the signal in glomus cells from the carotid body (37mmHg). Calcium signal evoked in astrocytes was reliably mimicked in chemical model of hypoxia (consumption of molecular oxygen by sodium dithionite (SDT) and not dependent on the activation of P2Y1 purinergic receptors, as shown for astrocytes pre-incubated with MRS2179 (a metabotropic ATP receptor antagonist).

The mitochondrion is the location where oxygen consumption ultimately takes place, for this reason we hypothesised that the sensor for astrocytic oxygen sensing resides in the

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mitochondria. Interestingly, in our simultaneous experiments of measurement of cytosolic calcium by fura-2 and $\Delta\psi_m$ by Rh123, each calcium spike was preceded well in advance (~30s or until mitochondrial depolarization reaches maximum drop for this condition) by a massive mitochondrial depolarization (~70%) which was reversible if episodes of hypoxia lasted no more than 2 min. Moreover, when we slightly depolarized mitochondria with 0.5 μM FCCP prior to induction of hypoxia, no more calcium signal could be evoked in astrocytes *in vitro* as well *in vivo* experiments. To confirm this finding, we have used PINK1 (PTEN-induced kinase 1; a Parkinson's disease related gene) KO mouse astrocytes with well-established reduction of mitochondrial membrane potential as a bioenergetic consequence of the mutation and main feature of the disease [59, 60]. As expected, no calcium signal could be evoked upon application of argon for 2 min (mild hypoxia). All this data point towards a certain threshold value for mitochondrial depolarization, that is necessary to be achieved at once, in order to trigger an astrocytic calcium response (Figure 2).

Delay in the rate of proton pumping by the enzymatic complexes of the electron transport chain will logically result in congestion of the electron transfer and thus in inhibition of mitochondrial respiration and consequently will enable the production of reactive oxygen species [15] by the collision of unpaired electrons with the surrounding molecules (lipids, proteins). In [58], we have unravelled the mechanism of oxygen sensing of astrocytes to engage as a major step the production of mitochondrial ROS. In experiments with MitoSOX we clearly show that superoxide is produced upon induction of mild hypoxia and the rate of superoxide production could be set back to basal by depolarizing the mitochondria with 0.5 μM FCCP or could be completely abolished by application of mitochondrial-targeted ROS scavenger MitoQ [61]. In fact, further step to lipid peroxidation is involved when the partial oxygen pressure $_PO_2$ was lowered, as measured in C11-BODIPY experiments in astrocytes. Again, the rate of lipid peroxidation remained at basal when astrocytes were pre-treated with 0.5 μM FCCP to slightly depolarize mitochondria prior to the hypoxia onset. Moreover, calcium signal that has been evoked by hypoxic condition in astrocytes have been completely abolished by the application of MitoQ or Vit.E, which affirms the finding that mitochondrial superoxide followed by lipid peroxidation are able to trigger calcium signalling event in response to altered atmospheric oxygen in astrocytes (Figure 2).

In this manner, we have been able to show the mechanism by which astrocytes are able to sense physiological changes in brain oxygenation and astrocytic mitochondria being the physical place of the CNS oxygen sensor.

Involvement of mitochondrial superoxide production in the mechanism of response to hypoxia and activation of the physical breathing can suggest possible negative effect of high doses of antioxidants (including mitochondrial scavengers) on the whole-body respiration during physical exercises or at high altitudes.

Conclusion remarks

For long time reactive oxygen species have been considered to be by-products of biological reactions that are highly aggressive and damaging to the surrounding tissue. It is necessary to

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accentuate on the fact that ROS are not cellular waste products, but actually signalling molecules that are very important for proper functioning of the organism, including proliferation, differentiation, aging, transcription factor regulation, inflammation, and other regulatory functions.

There are no doubts that mitochondrial ROS are involved in the mechanism of acute (i.ie. anoxia) or chronic pathology (i.e. neurodegeneration) but because of important implication of mitochondrial superoxide and hydrogen peroxide in physiology, more accurately developed strategy for treatment with ROS scavengers and mitochondrially targeted antioxidants should be used.

Moreover, it should also be pointed out that lipid peroxidation which has always been accepted as a marker for late oxidative stress has an essential role as a source of substrates and as an activator of phospholipases, thus playing an important role in cells housekeeping and signalling.

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Figure Legend

Figure 1. Schematic representation of the major mitochondrial ROS producers and targets.

Figure 2. Mechanism of oxygen sensing in astrocytes. Reduction of the partial $_PO_2$ in the medium leads to inhibition of ETC in mitochondria. It stimulates production of superoxide followed by lipid peroxidation. Oxidised lipids activate phospholipase C, which uses them for IP_3 production. IP_3 releases Ca^{2+} from the endoplasmic reticulum which triggers signalling cascade in astrocytes.

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Figures caption:

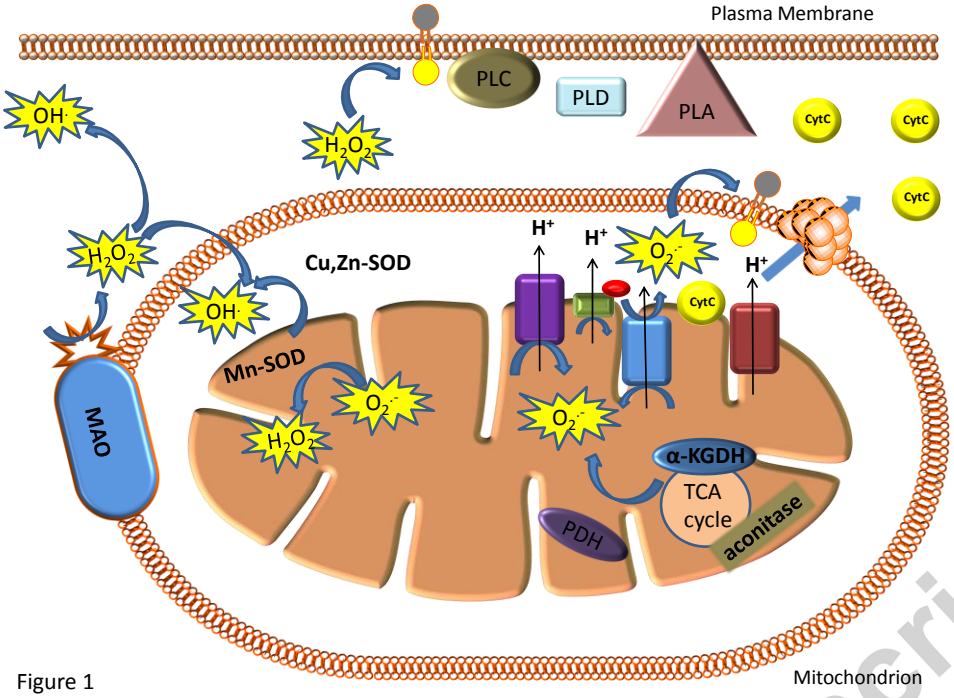


Figure 1

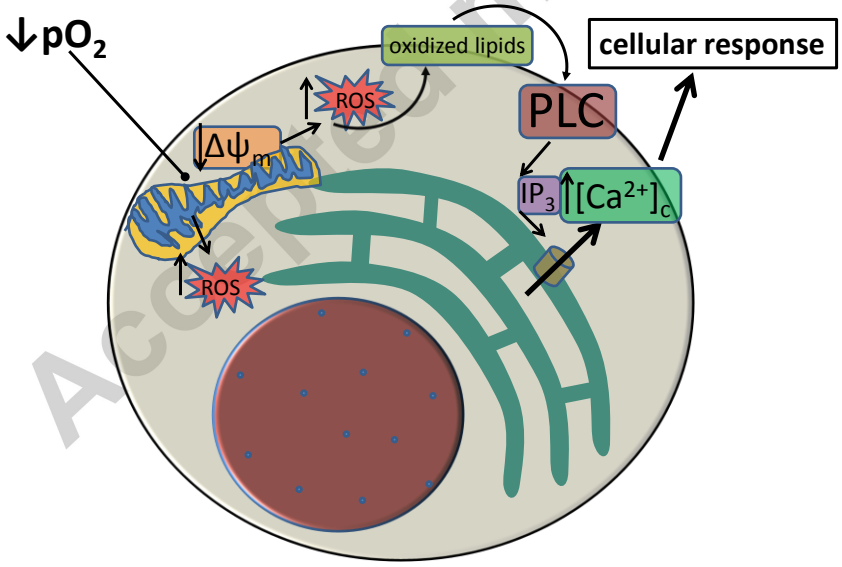
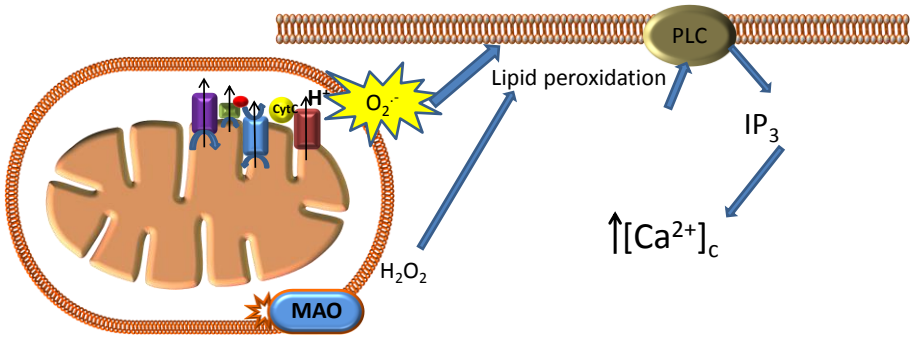


Figure 2

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ACCEPTED MANUSCRIPT



Graphical abstract

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Highlights

1. Mitochondria produce and modify ROS in response to physical and chemical stimuli.
2. Mitochondria are able to sense changes in atmospheric O_2 and produce ROS to hypoxia
3. Mitochondrial ROS can induce lipid peroxidation and activate PLC-induced Ca^{2+} signal
4. Superoxide from mitochondria can modify activity of ion channels and receptors